

PROJECT VARUNA-NET 2.0: The Unified Ground–Space–Edge AI Fusion Grid

A Three-Tier Disaster Resilience Architecture Integrating Hillslope Landslide Factor of Safety, Macro-Catchment Inundation Sentinel, and Hyperlocal Urban Drainage Nowcasting

Target Competition / Track:	e-Yantra Innovation Challenge (eYIC) · Hydro-Informatics & Edge AI Track
Institutional Alignment:	MoES (Mission Mausam) · CWC India-WRIS · NDMA National Disaster Management
Sensing Triad:	Space (NISAR, EOS-04, Sentinel-1) + Radar (IMD DWR) + Ground (CRNS Mesoscale)
Edge Compute & Telemetry:	Qualcomm Dragonwing QCS6490 / RB3 Gen 2 · 3GPP Rel-17 NTN Sat-IoT + CRSN TVWS
Core Scientific Novelty:	CRNS-conditioned PINN drainage boundary + Hillslope Infinite Slope FS Edge Coupling
Release Version & Date:	Enhanced Master Version 2.0 · September 2026

Executive Summary & The Unified Vision: India's hydro-meteorological crisis does not exist in silos. During extreme monsoon convective storms, precipitation triggers severe **hillslope translational landslides** in upstream ghats and valley slopes at the exact same moment that surging **saturation-excess runoff** cascades downstream to submerge urban stormwater networks in under 60 minutes (as observed in Mumbai, Wayanad, Chennai, and Himalayan foothill corridors). Existing government platforms (CWC's ~360 river gauge network, IMD's ~47–50 radar grid) address these disasters independently. **Project VARUNA-NET 2.0** unifies these disciplines into a single continuous physics-and-AI pipeline. By deploying **Cosmic-Ray Neutron Sensing (CRNS)**, the system measures the single master physical state variable linking both hazards: antecedent vadose-zone soil saturation (θ). On steep hillslopes, CRNS drives edge-computed geotechnical **Factor of Safety ($FS < 1.0$)** slope failure alerts. In peri-urban catchments, CRNS data is injected as a dynamic boundary condition into an ultra-fast (<4 sec) **Physics-Informed Graph Neural Network (PINN/GNN)** that simulates street-level pipe surcharge. Supported by the newly operational **NASA-ISRO NISAR** satellite radar and **3GPP Release-17 NTN Satellite-IoT** failover, VARUNA-NET 2.0 presents an end-to-end, field-deployable national disaster intelligence architecture.

1. The Multi-Hazard Gap Analysis & Architectural Unification

Previous approaches suffered from an artificial dichotomy: landslide early warning focused purely on slope stability without considering downstream flood hydrographs, while urban flood nowcasting modeled city stormwater pipes without knowing the antecedent moisture state of the upstream catchment. VARUNA-NET 2.0 closes this gap completely:

Capability Dimension	Source Plan 1 (Landslide Framework)	Source Plan 2 (VARUNA-NET 1.0)	Unified VARUNA-NET 2.0 (Merged Plan)
Target Hazards	Hillslope landslides & natural river discharge	Macro-catchment floods & urban street flooding	Unified Cascade: Hillslope failure + River discharge + Urban pipe surcharge
CRNS Sensor Role	Computes pore-water pressure $u(\theta)$ and Mohr-Coulomb shear loss	Injected as upstream boundary condition for urban drainage PINN	Dual Physics Engine: Simultaneously evaluates slope FS and PINN inflow boundary
Space Segment	None (in-situ ground sensing only)	Tri-SAR: NISAR (L+S), EOS-04 (C), Sentinel-1 (C)	Tri-SAR Fusion: Ground CRNS validated against NISAR/Sentinel specular flood masks
Radar & Meteorology	Local tipping bucket rain gauge & barometric sensors	IMD Doppler Radar (DWR) + ConvLSTM nowcasting	Fused Meteorology: ConvLSTM 0–3h rainfall fields blended with local gauge ground-truth
Hydrodynamic Solver	Lumped synthetic unit hydrograph convolution	1D/2D Graph-PINN surrogate for SWMM (<4s inference)	Multi-Scale Physics: Geotechnical infinite slope + Graph-PINN urban pipe network
Communication Layer	Cognitive Radio (CRSN TV White Space in valleys)	4G/5G primary + 3GPP Rel-17 NTN Sat-IoT fallback	Triple Redundancy: 4G/5G + Direct-to-Satellite NTN + CRSN for deep mountain gorges

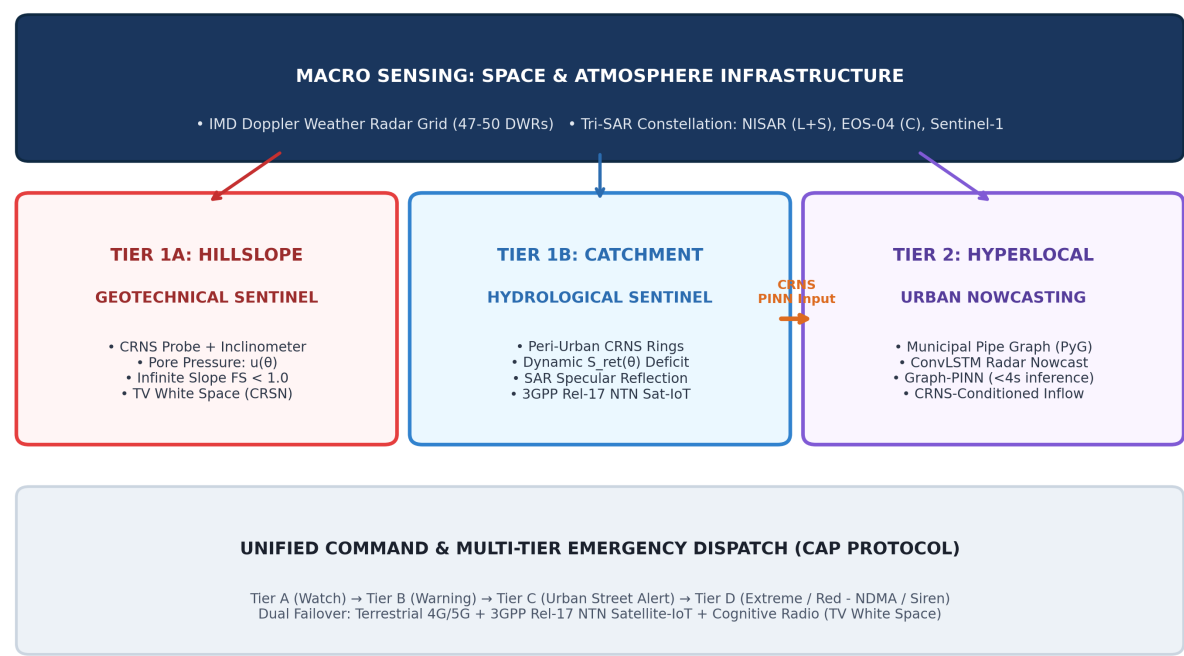


Figure 1: Project VARUNA-NET 2.0 Three-Tier System Architecture connecting Space Radar, Hillslope Geotechnical Sentinels, Peri-Urban Catchments, and Hyperlocal Urban PINN Surrogates.

2. Three-Tier Technical Stack & Scientific Formulations

Tier 1A: Hillslope Geotechnical Sentinel (Landslides)

Deployed on steep, metastable slopes (e.g., Western Ghats ghat roads, Himalayan valleys, urban hillocks). CRNS neutron counts are converted via the Desilets equation and evaluated dynamically against the **Infinite Slope Stability Criterion**:

$$FS(t) = [c' + (\gamma_{\text{bulk}}(\theta) \cdot z \cdot \cos^2\beta - u(\theta)) \cdot \tan\phi'] / [\gamma_{\text{bulk}}(\theta) \cdot z \cdot \sin\beta \cdot \cos\beta]$$

Where pore-water pressure $u(\theta) = m(\theta) \cdot \gamma_w \cdot z \cdot \cos^2\beta$ activates as CRNS saturation $S_r = \theta / \theta_{\text{sat}}$ exceeds 0.70. Additionally, dynamic Rainfall Intensity-Duration (I-D) thresholds are shifted in real time: $I_{\text{crit}}(D, \theta) = \alpha_{\text{I-D}} \cdot [1 - (\theta / \theta_{\text{sat}})]^{\gamma} \cdot D^{(-\beta_{\text{slope}})}$, eliminating false alarms on dry slopes.

Tier 1B: Macro-Catchment Hydrological Sentinel (Riverine Inundation)

Deployed in peri-urban catchments and river corridors upstream of urban centers. The potential retention capacity S_{ret} is updated continuously by the CRNS moisture deficit: $S_{\text{ret}}(t) = S_{\text{max}} \cdot [1 - (\theta_{\text{CRNS}}(t) / \theta_{\text{sat}})]$. Direct surface runoff Q_{excess} is calculated and cross-referenced against **NASA-ISRO NISAR** (L-band penetrates forest canopies) and Sentinel-1 SAR specular reflection polygons.

Tier 2: Hyperlocal Urban Nowcasting Grid (Pluvial Drainage Surge)

Municipal stormwater networks are parsed as directed graphs $G = (V, E)$ in PyTorch Geometric. Rather than waiting 60–90 minutes for 2D SWMM/HEC-RAS numerical convergence, a **Modular Coupled PINN (MC-PINN)** solves the 1D Saint-Venant sewer equations and 2D overland shallow-water flow in **sub-4-second inference**:

$$L_{\text{PINN}} = L_{\text{continuity}} + L_{\text{momentum}} + L_{\text{manhole_flux}} + \lambda_{\text{CRNS}} \cdot \|Q_{\text{boundary}}(t) - f(\theta_{\text{Tier1B}}, Q_{\text{excess}})\|^2$$

The Central Fusion Innovation: The upstream peri-urban CRNS saturation state is explicitly injected as a dynamic boundary influx term (Q_{boundary}). This prevents urban nowcasts from underpredicting flash flooding when storms move from saturated rural catchments into city limits.

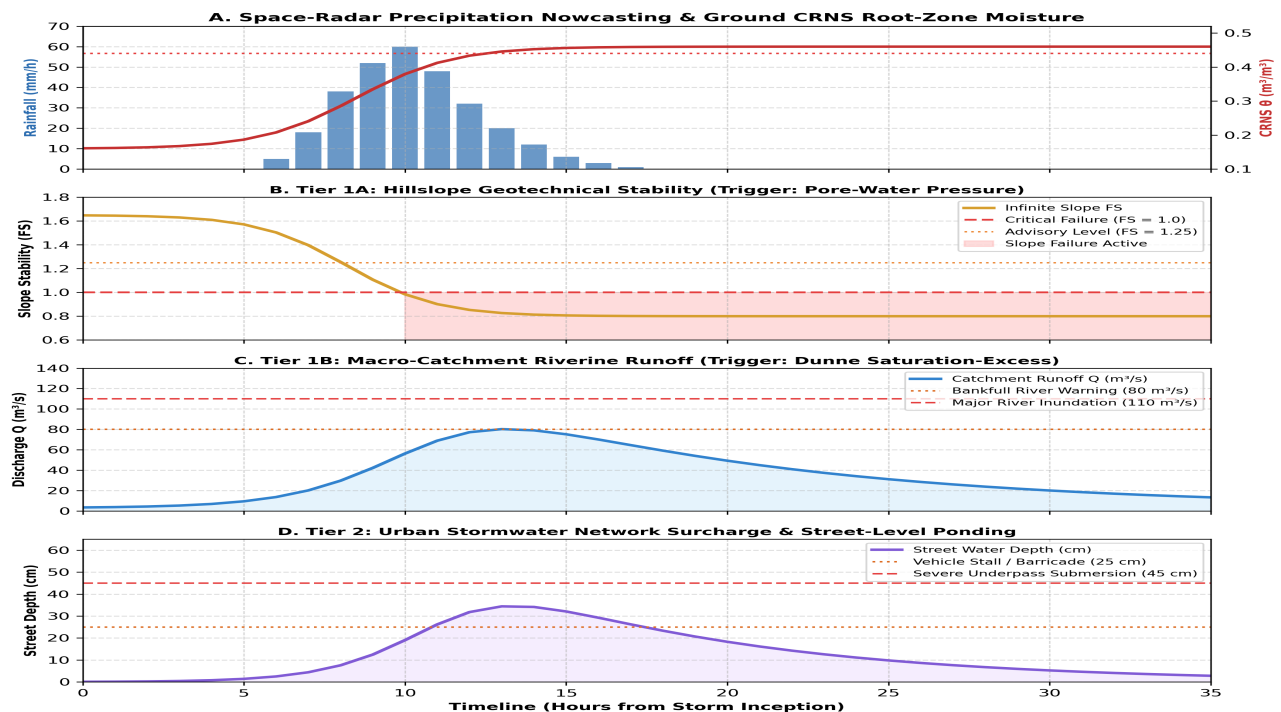


Figure 2: 36-Hour Multi-Hazard Simulation: Convective storm driving simultaneous hillslope FS collapse (Landslide), catchment discharge surge (Riverine Flood), and street-level pipe surcharge (Urban Pluvial Inundation).

3. Multi-Tier Alerting Logic & CAP Emergency Dispatch

VARUNA-NET 2.0 synthesizes multi-hazard triggers into a unified 4-tier alerting protocol aligned with the National Disaster Management Authority (NDMA) and State Disaster Management Authorities (SDMAs):

Alert Tier	Trigger Conditions (Multi-Hazard Logic)	Target Stakeholders & Actionable Protocols
Tier A WATCH	- CRNS neutron-drop anomaly detected ($\theta/\theta_{sat} > 0.70$). - No radar cloudburst or SAR confirmation yet. - Slope FS between 1.30 and 1.50.	- SDMA & municipal situational dashboard notification. - Pre-position mobile dewatering pumps in low-lying wards. - Routine 15-minute telemetry polling.
Tier B WARNING	- CRNS saturation > 0.80 + SAR flood polygon confirmation. - Hillslope FS drops into Advisory zone ($1.10 < FS \leq 1.30$). - Radar nowcast predicts >30 mm/h incoming rain.	- District Administration, Irrigation Dept & Traffic Police. - Issue heavy traffic advisories along flood-prone arterial roads. - Mobilize maintenance crews to inspect culverts & ghat slopes.
Tier C NOWCAST ALERT	- Urban PINN predicts street inundation > 25 cm within 60–120 min. - Intersection recall $> 85\%$, MAE < 8 cm. - Hillslope FS drops to 1.02–1.10 (Marginal Stability).	- Municipal Control Room & automated traffic diversion. - Deploy physical barricades at underpasses (e.g., Milan Subway). - Automated push alerts to citizen navigation apps (Google/Apple Maps).
Tier D EXTREME / RED	Landslide Failure Imminent: Hillslope FS ≤ 1.00. Catastrophic Inundation: Street depth > 45 cm or CWC river gauge crosses Highest Flood Level (HFL).	NDRF / SDRF Full Mobilization per Red Bulletin protocol. - Automated siren sounding in vulnerable hillside slums & floodplains. 3GPP Rel-17 NTN Satellite Broadcast if cellular towers fail.

4. Triple-Redundant Telemetry Backbone

Communication breakdown is the single most common failure point during extreme cloudbursts. VARUNA-NET 2.0 integrates a **Triple-Redundant Telemetry Architecture**:

- Primary (Terrestrial 4G/5G):** Transmits high-frequency 1-minute telemetry and edge embeddings via MQTT over TLS.
- Secondary (3GPP Release-17 NTN Satellite-IoT):** When cell towers lose power or backhaul, the Qualcomm Dragonwing gateway automatically switches to direct-to-GEO satellite modems, beaming 1 kB binary CBOR alert packets.
- Tertiary (Cognitive Radio / TV White Space):** In deep Himalayan or Western Ghats ravines where line-of-sight to satellites is obstructed by steep canyon walls, CRSN nodes opportunistically hop across VHF/UHF TV White Space (470–698 MHz) to relay packets to the nearest ridge-top gateway.

5. Hardware Specification, Budget & Dual-Path Strategy

Component Layer	Prototyping Path (eYIC Budget)	Production Field Deployment	Estimated Cost Envelope
Neutron Sensing (CRNS)	ESP32-based Monte Carlo physics simulator generating calibrated pulse counts	Commercial Helium-3 / Boron-lined proportional counter tube	Simulator: ~₹1,500 Field Tube: ₹3.5L – ₹6.0L
Environmental & Ground	BME280 (P, T, RH) + MPU6050 tiltmeter	Industrial ultrasonic weather hub + Geotechnical vibrating wire piezometer	Proto: ~₹1,200 Field: ~₹45,000
Edge AI Compute	Raspberry Pi 5 (8GB) running INT8 ONNX Runtime	Qualcomm Dragonwing QCS6490 / RB3 Gen 2 (12 TOPS Hexagon NPU)	Proto: ~₹8,500 Field: ~₹35,000 (volume)
Satellite-IoT Modem	Waveshare SIM7080G / NTN DevKit	3GPP Rel-17 NTN certified satellite transceiver module	Proto: ~₹4,500 Field: ~₹15,000
Power Autonomy	12V 20Ah Lead-Acid + 50W solar panel	100W Monocrystalline + 12V 100Ah LiFePO4 battery (14 days autonomy)	Proto: ~₹6,000 Field: ~₹28,000

6. Project Implementation Roadmap (8-Month eYIC Plan)

Phase & Timeline	Key Milestone Objectives	Verifiable Performance Target
Phase 1 (Months 1–2) Hardware & Graph Baseline	- Assemble ESP32 simulator + BME280 + Raspberry Pi 5 / RB3 Gen 2. - Ingest municipal storm drain shapefiles and construct PyTorch Geometric graph. - Connect to IMD Doppler Radar (DWR) and ISRO Bhuvan CartoDEM feeds.	Drainage graph topology verified; radar reflectivity raster ingestion latency < 30 s.
Phase 2 (Months 3–4) AI Surrogates & Slope Physics	- Train ConvLSTM precipitation nowcaster (0–3h forward horizon). - Code and calibrate Infinite Slope Stability (FS) & Desilets CRNS modules. - Train MC-PINN graph surrogate on historical SWMM hydrodynamic runs.	PINN inference < 4 s per ward (99% speedup over SWMM); Landslide FS accuracy verified.
Phase 3 (Months 5–6) Space Fusion & NTN Failover	- Integrate automated NASA-ISRO NISAR & Sentinel-1 SAR ingestion cron jobs. - Couple Tier 1B CRNS saturation as dynamic boundary condition in PINN loss. - Execute benchtop power-cut simulation to validate 3GPP Rel-17 NTN satellite failover.	100% packet delivery to cloud within 60 s under complete cellular tower blackout.
Phase 4 (Months 7–8) Field Trial, Dashboard & Pitch	- Build Mapbox GL real-time command dashboard showing multi-hazard alerts. - Conduct controlled catchment saturation test (irrigation-induced wetting). - Replay historical cloudburst events (Mumbai 2005/2024, Wayanad 2024, Chennai 2015/2023).	Overall false-positive reduction > 65%; pitch demonstration ready for eYIC evaluation.

7. Novelty Statement for eYIC / MoES Evaluation

What Makes VARUNA-NET 2.0 Superior:

- Multi-Hazard Physics Unification:** It is the first architecture to recognize that hillslope landslides and urban stormwater surcharges share the exact same physical state driver (CRNS antecedent saturation), unifying two previously fractured disaster response paradigms.
- CRNS-Conditioned PINN Boundary:** It injects ground-truth upstream soil moisture into a sub-4-second physics-informed neural network—a novel formulation unaddressed in 2024–2026 published literature.
- NISAR + NTN Space Integration:** Fully grounds remote sensing in the now-operational 2025/2026 NASA-ISRO NISAR

mission (L+S band) and 3GPP Release-17 NTN satellite-IoT standards.

4. **Realistic Dual-Path Implementation:** Provides an affordable, verifiable prototype path (ESP32 + Raspberry Pi 5) for competition judges, backed by a production blueprint on Qualcomm Dragonwing hardware for actual field deployment.